MELISSA ADU-POKU

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Full stack software engineer with 7+ years of experience designing and building scalable applications across frontend and backend systems, including incorporating AI-driven workflows to improve engineering efficiency. Strong background in React, TypeScript, Django, and Java. Experienced in building reusable UI systems, implementing robust backend services, integrating REST and GraphQL APIs, and maintaining high test coverage across the stack. Proven track record of improving system performance and delivering production-grade features.

SKILLS

Frontend: React | Next.js | React Native | TypeScript | JavaScript | HTML | CSS
Backend: Python | Django | Java | Node.js | API Design (REST, GraphQL)
Databases: MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling
Infrastructure & DevOps: Git | Docker | CI/CD | GitHub Actions | Kubernetes (collaboration)
AI Tools: Lovable | Cursor | Claude Code | Base 44 | GitHub Copilot | Replit
Development Practices: System Design | Agile/Scrum | Code Reviews | Component Libraries

PROFESSIONAL EXPERIENCE

MINERALTREE, INC., Cambridge, MA. **October 2018 – June 2025**

Senior Software Engineer, January 2024 – June 2025

Co-led the modernization of a legacy monolithic application, partnering across frontend and backend teams to migrate the UI to React and Redux, while aligning with evolving backend contracts, thereby improving maintainability, scalability, and reducing production UI bugs by 45%.

- Built a reusable design system and component library used across product engineering teams, cutting new feature UI delivery time by 40% and ensuring uniformity across applications.
- Collaborated on API design and integrated applications with 20+ REST and GraphQL microservices, supporting thousands of daily API requests with optimized client-side caching and error handling.
- Developed a comprehensive testing strategy with high-coverage unit and integration tests across hundreds of components, business-critical workflows, and API integrations, and built CI/CD pipelines with GitHub Actions for test automation and deployments, achieving high code and branch coverage across critical paths, and reducing regression-related incidents by 60%.
- Played a key role in sprint planning, on-call production support, and architectural reviews to strengthen team output and engineering quality.

Software Engineer, January 2020 – January 2024

Co-led the development of customer-facing features across the full stack, including dynamic dashboards and complex form workflows for internal operations and external customers.

- Designed frontend state management strategies aligned with backend data models to reduce redundant requests and improve perceived performance.
- Participated in backend integration work, debugging service-to-service communication issues and validating API contracts.
- Implemented new full stack features and improved application observability by enhancing structured logging and monitoring practices in collaboration with backend teams.
- Developed and maintained testing pipelines spanning unit, integration and end-to-end coverage.

Software Intern, October 2018 – January 2020

Developed a Django/Python-based financial reporting and payments monitoring platform used internally across Operations and Finance teams to manage multi-million dollar monthly transaction volumes.

- Designed and implemented REST API endpoints to support reporting workflows and integrations with Salesforce and other tools, reducing churn-related operational friction.
- Optimized complex MySQL queries supporting high traffic financial reports, cutting average query time by ~55% and improving MineralTree application page load performance.
- Implemented downloadable financial reports to automate recurring reporting processes for customers.

N12 TECHNOLOGIES, INC., Cambridge, MA. **August 2016 – June 2017**

Business Analyst

Automated sales fulfillment workflows using Salesforce + Zapier, reducing order processing time from hours to minutes.

EDUCATION

Bachelor of Arts (B.A.), Economics Smith College, Northampton, MA **| 2015**